Angles & 2D Shapes

Year 4 · **Homework Worksheet**

Worked Example

Worked example: What type of angle is 90°?

Answer: An angle of exactly 90° is called a right angle.

Easy / Standard (15 questions)

1. What type of angle is 20°?
2. What type of angle is 150°?
3. Which of these is a right angle? A) 60° B) 90° C) 120° D) 170°
4. What type of angle is exactly 180°?
5. How many sides does a hexagon have?
6. How many sides does a pentagon have?
7. Which shape has 4 sides but not all equal, with 4 right angles? A) square B) rectangle C) triangle D) pentagon
8. How many sides does an octagon have?
9. Select ALL of these angles that are OBTUSE (between 90° and 180°): A) 45° B) 120° C) 95° D) 30° E) 160°
10. How many sides does a triangle have?
11. What is the name for an angle that is exactly 90°?
12. Which shape has 5 sides? A) square B) pentagon C) hexagon D) triangle
13. How many lines of symmetry does a rectangle (non-square) have?
14. What type of angle is 89°?
15. How many sides does a square have?

Standard (15 questions)

1. What type of triangle has one angle greater than 90°?
2. Which triangle has all three angles equal (all 60°)? A) scalene B) isosceles C) equilateral D) right- angled
3. What type of triangle has one right angle?
4. How many lines of symmetry does a square have?
5. Angles on a straight line always add up to how many degrees?
6. Which of these describes an isosceles triangle? A) exactly two sides equal B) all sides equal C) no sides equal D) all angles different
7. Angles around a point always add up to how many degrees?
8. Select ALL of the following that always have 4 right angles: A) square B) rectangle C) rhombus (non-square) D) parallelogram (non-rectangle) E) any quadrilateral
9. How many lines of symmetry does a regular pentagon have?

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10. What is the sum of the interior angles of a triangle?

11. Which of these shapes has exactly 1 curved side? A) semicircle B) square C) rectangle D) hexagon

12. A right angle is split into two smaller angles. If one is 40°, what is the other?

13. How many sides does a nonagon have?

14. Which shape has 4 equal sides but its angles are not necessarily 90°? A) rhombus B) rectangle C) square D) triangle

15. How many right angles does a square have?

Standard / Advanced (15 questions)

1. Take a right angle and add 100° to it. What type of angle results, and what is its size in degrees?

2. A straight line is split between two angles. One of them measures 125° — what size must the other one be? A) 45° B) 55° C) 65° D) 75°

3. Within a triangle, two angles each measure 60°. Whatever is left over from 180° belongs to the third angle — what size is it?

4. Picture a shape with 4 equal sides but no right angles anywhere. Select ALL of the statements below that are TRUE about this shape: A) It could well be a rhombus B) It has to be a square C) It counts as a quadrilateral D) It could never be a parallelogram E) It has exactly 4 sides

5. Starting from a 40° angle and doubling it gives a new angle. What type of angle is this new one?

6. A straight line holds two angles together, one of 65° and the other labelled x°. Find the value of x.

7. Around a single point sit three angles measuring 120°, 100° and x°. What is x?

8. Three angles of a quadrilateral measure 95°, 85° and 90°. Work out the size of the fourth angle. A) 80° B) 85° C) 90° D) 95°

9. One base angle of an isosceles triangle measures 65°. What sizes are the other two angles in the triangle?

10. When two straight lines cross over one another, vertically opposite angles are formed that are always equal. Select ALL of the pairs below that fit this description: A) 50° and 50° B) 50° and 130° C) 70° and 70° D) 90° and 90° E) 100° and 80°

11. Three of a quadrilateral's four angles happen to be right angles. What size must the remaining angle be?

12. A straight line is divided into two angles, and one of them is 40° larger than the other. What sizes must the two angles be?

13. A triangle's angles include one of 90° and another of 35°. What size is left for the third angle? A) 45° B) 55° C) 65° D) 75°

14. A regular hexagon's interior angles add up to a total of 720°. Since all its angles are equal, what size is each one?

15. Instead of just two, a straight line here is divided into three separate angles: 50°, 60° and x°. What value does x take?

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